O-RAN AI/ML Workflow

Architecture, Interfaces, Deployment Scenarios & Research Gaps

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: Polese, Bonati, D'Oro, Basagni, Melodia, *"Understanding O-RAN: Architecture, Interfaces, Algorithms, Security, and Research Challenges"* (arXiv:2202.01032, 2022). **AI/ML workflow** , **(research gap)** .

1 : "AI/ML workflow"

O-RAN (disaggregation) , RAN ML . O-RAN " " **"ML "** . . .

.

1. AI/ML workflow ?
2. **(SMO / Non-RT RIC / Near-RT RIC / RAN node)** , ?
   (timescale) , .

2 : AI/ML O-RAN

AI/ML workflow , “ ” .

1: O-RAN

SMO / Non-RT RIC	>1 s	rApp	A1, O1, O2	· , , intent
Near-RT RIC	10 ms – 1 s	xApp	E2	RRM ,
RAN node (CU/DU)	<10 ms	(embedded)	—	· HARQ real-time

(nested control loop) , . T_ℓ

$$T_{\text{Non-RT}} > 1\,\text{s} \gg T_{\text{Near-RT}} \in [10\,\text{ms}, 1\,\text{s}] \gg T_{\text{RT}} < 10\,\text{ms}$$

. **<10 ms real-time** **(further study)** , .

3 AI/ML Workflow

O-RAN WG2 AI/ML , .

1. **Data Preparation** — · · feature engineering
2. **AI/ML Training** — ( )
3. **Validation & Deployment** —
4. **Inference** —
5. **Continuous Operations** — ·

.

- : . fine-tuning .
- : **RIC** , CU/DU (further study) .

4 Workflow : (Fig. 14)

Fig. 14 5 (Model Management / Continuous Operations / Data Preparation / Training / Inference) 5 . .

4.1

2: O-RAN AI/ML (Fig. 14)

	Model Mgmt	Continuous Ops	Data Prep	Training	Inference
1.1	Non-RT	Non-RT	Non-RT	Non-RT	Non-RT
1.2	SMO	Non-RT	Non-RT / Near-RT	Non-RT	**Near-RT**
1.3	SMO	Non-RT	SMO	SMO	Non-RT
1.4	Non-RT	**Near-RT**	Non-RT	Non-RT / SMO	**Near-RT**
1.5 (FFS)	Non-RT	( )	( )	( )	**O-CU / O-DU**

4.2

- **1.1** — : Non-RT RIC . >1 s · . .
- **1.2** — / : Non-RT( ), **Near-RT** 10 ms–1 s . .
- **1.3** — : Model Management Training **SMO** , · Non-RT . SMO Non-RT .
- **1.4** — : 1.2 **Continuous Operations Near-RT** . self-monitor Non-RT . RIC .
- **1.5 (FFS)** — **RAN** : **O-CU/O-DU** . TTI real-time , **(For Further Study)** .

: 1.5 SMO · Near-RT RIC " " **FFS** . FCAPS · RRM .

5 Chained AI/ML Access Platform

(chaining) . , access platform( ) / .

( ):

$$\underbrace{\text{rApp:}}_{A} \rightarrow \underbrace{\text{xApp Y:}}_{C} \rightarrow \underbrace{\text{xApp Z:}}_{D=f(A,B_{\mathrm{KPM}},C)}$$

( ) ( ) , hierarchical .

6 ML Use Case

ML , .

- **Resource Allocation** (·) — ML . (ColO-RAN, NexRAN) .
- **Mobility** — · .
- **QoE/QoS** — .
- **RAN Sharing / Massive MIMO /** (UAV · · IoT).

, **1.5( )** .

7

AI/ML workflow / . **1.5( )** .

7.1 1 — Real-time(<10 ms)

RIC(10 ms) , **TTI** (, link adaptation) ML . , τ_{inf} τ_{sched} TTI .

$$\tau_{\mathrm{inf}} + \tau_{\mathrm{sched}} \leq \text{TTI} \quad (\text{TTI} \approx 0.5\text{–}1\,\text{ms})$$

· , 1.5 FFS .

7.2 2 — (model delivery)

“Model Management Non-RT, Inference O-DU” 1.5 **weights Non-RT** .
.

3:

A1	· enrichment
E2	RRM ( )
O1	FCAPS

7.3 3 — (resource isolation)

RAN ML workload . (i) GPU/CPU , (ii) RAN (jitter), (iii) QoS
. **RAN workload AI inference workload** · .

7.4 4 — Cross-layer

(: — entropy, logprob, attention) **MAC utility** cross-layer · .
1 (+).

8 :

1.5 , **(hierarchical control)** .

```
Non-RT RIC  →      /   prior      (> 1 s)
      A1
Near-RT RIC →  ·   /            (10 ms-1 s)
      E2
O-DU (cuMAC) →  +      PRB          (< 10 ms)
```

- **(RIC)** · · .
- ( ) TTI + .

— + — , **FFS** **1.5** . (,) Near-RT ,
(GH200 / pyAerial / cuMAC / ROS2) 1 · 3 .

4:

1 (real-time)	sub-ms feasibility	GH200 + pyAerial
3 ( )	RAN+AI workload	Aerial
4 (cross-layer)	→ utility	cuMAC
	.	Near-RT

9

O-RAN AI/ML workflow 5 4 , . **RAN** **1.5** **TTI**
FFS , cross-layer · systems-level .